MRHevo: statistical model and methods

Statistical model for individual-level genotype-outcome data

- X exposure
- Y outcome
- $\boldsymbol{Z}$ vector of genotypic instruments, of length equal to the number J of unlinked loci
- $\boldsymbol{\alpha}$ vector of coefficients of effects of instruments Z on exposure X
- $\boldsymbol{\beta}$ vector of coefficients of direct (pleiotropic) effects of instruments on outcome Y
- $\boldsymbol{X_u}$ unpenalized covariates
- θ parameter for causal effect of X on Y
- $\beta 0$, $\boldsymbol{\beta_u}$ parameters for intercept and unpenalized covariates $\boldsymbol{X_u}$

We have a dataset of N individuals with measurements of the outcome Y and the genetic instruments $\boldsymbol{Z}$. From summary statistics we have estimates $\hat{\boldsymbol{\alpha}}$ of the effects $\boldsymbol{\alpha}$ of the instrument on the exposure, with corresponding standard errors $s_{\alpha(1)}, \ldots, s_{\alpha(J)}$.

We specify a Bayesian full probability model as below

$$\alpha_j \sim \mathcal{N}\left(\hat{\alpha}_j, s^2_{\alpha(j)}\right)$$

$$\mathbb{E}\langle X \rangle = \alpha_0 + \boldsymbol{Z\alpha}$$

$$g\left(\mathbb{E}\langle Y \rangle\right) = \beta_0 + \boldsymbol{X_u\beta_u} + \boldsymbol{Z\beta} + \theta\mathbb{E}\langle X \rangle$$

where $g\left(\right)$ is a link function.

To calculate the likelihood as a function of the causal effect parameter θ, we have to marginalize over the distribution of the direct effects $\boldsymbol{\beta}$ given the data

As the form of the distribution of pleiotropic effects over loci is unknown, any realistic statistical model has to specify a prior on these effects that encompassses a broad family of symmetric distributions ranging from a spike-and-slab mixture to a Gaussian. A convenient model for the direct effects is the horseshoe prior, which is equivalent to a mixture of spike-and-slab distributions (Carvalho, Polson, and Scott 2009). This eliminates the computational inconvenience of having to average over all possible partitions of the variables into two disjoint sets (spike, slab) as in the contamination mixture model (Burgess et al. 2020) or over different settings of the number of nonzero effect parameters (Xue, Shen, and Pan 2021). The horseshoe prior specifies a global shrinkage parameter τ and J local shrinkage parameters λ_j. The regularized horseshoe prior modifies this by specifying a global regularization parameter η that controls the size of the largest nonzero effects.

$$\beta_j \sim \mathcal{N}\left(0, \tau^2\tilde{\lambda}_j^2\right), \tilde{\lambda}_j^2 = \frac{\eta^2\lambda_j^2}{\eta^2 + \tau^2\lambda_j^2}$$

Half-Cauchy priors are specified on the unregularized local scale parameters λ_j.

$$\lambda_j \sim C^+(0,1)$$

A weakly informative prior is specified for the regularization parameter η^2:

$$\eta^2 \sim \text{Inverse-Gamma}\left(\frac{\nu_{\text{slab}}}{2}, \frac{\nu_{\text{slab}} s_{\text{slab}}^2}{2}\right)$$

This translates to a Student t prior with ν_{slab} degrees of freedom and scale s_{slab} on the largest direct effects.

The horseshoe prior is a multivariate scale mixture of normals: even though there is one local scale parameter for each effect size parameter, these local scale parameters can be integrated out so that the effect parameters are still identifiable (Carvalho, Polson, and Scott 2009). The heavy tail of the half-Cauchy priors on λ_j allows some of the regression coefficients to escape the shrinkage imposed by small values of the global parameter τ. These nonzero coefficients are the slab component of the spike-and-slab distribution. The value specified for the scale of the global shrinkage parameter τ encodes an expectation of the fraction of coefficients that are nonzero.

The heavy tail of the half-Cauchy distribution allows some of the regression coefficients to escape the shrinkage imposed by the global parameter τ. The regularization parameter η regularizes the scale of the nonzero coefficients (those that are in the slab of the spike and slab distribution). Even the largest coefficients will be regularized as a Gaussian with variance η^2.

The value of ν_{slab} (set in `MRHevo` with the argument `slab_df`) controls the shape of the prior distribution of η. Piironen and Vehtari recommend setting $\nu_{\text{slab}} = 1$ which is equivalent to a Cauchy prior on η. This allows η to take very large values, which is not realistic in the setting of polygenic effects on a complex trait. It also leads to awkward posterior geometry: in the region of the posterior where η is large and τ is very small, funnelling occurs. This is evident as divergent transitions with the NUTS sampler. By default, MRHevo sets $\nu_{\text{slab}} = 2$. Larger values may be required to regularize the sampler so that it does not draw very large values of η.

The scaling factor s_{slab} (set in `MRHevo` with the argument `slab_scale`) is specified based on prior information about the size of the largest direct effects. This information will usually be available from genome-wide association studies of the outcome.

A half-t distribution is chosen for the global scale parameter τ

$$\tau \sim t^+(0, s_{\text{global}}, \nu_{\text{global}})$$

Specifying $\nu_{\text{global}} = 1$ gives a half-Cauchy prior.

The scaling factor s_{global} is not set directly by the user. Instead the user specifies a prior guess f_0 (set in `MRHevo` with the argument `fraction_pleio`) about the fraction of nonzero direct effect coefficients. This setting depends upon $\mathcal{I}$, the Fisher information about the direct effect coefficient β_j, which in turn depends on the size of the outcome dataset. MRHevo calculates the setting of s_{global} from the Fisher information and the value of f_0 specified by the user, so that that the median τ_0 of the prior on τ is the value at which the fraction of nonzero coefficients is f_0.

Encoding the prior expectation of the effective fraction of nonzero effects

For a coefficient β_j with prior $\beta_j \sim \mathcal{N}(0, \tau^2\lambda_j^2)$ and Gaussian likelihood with Fisher information $\mathcal{I}_j$, the posterior mean is a weighted average of zero (the prior mean) and the maximum likelihood estimate $\hat{\beta}_j$:

$$E[\beta_j \mid \hat{\beta}_j] = \frac{\tau^2\lambda_j^2}{\tau^2\lambda_j^2 + \mathcal{I}_j^{-1}}\hat{\beta}_j = (1-\kappa_j)\hat{\beta}_j$$

The shrinkage coefficient κ_j is the fraction by which the prior shrinks the posterior mean towards zero:

$$\kappa_j = \frac{\mathcal{I}_j^{-1}}{\tau^2\lambda_j^2 + \mathcal{I}_j^{-1}} = \frac{1}{1+\tau^2\lambda_j^2\mathcal{I}_j}$$

For the regularized horseshoe, λ_j is replaced by the regularized local scale $\tilde{\lambda}_j$:

$$\kappa_j = \frac{1}{1+\tau^2\tilde{\lambda}_j^2\mathcal{I}_j}$$

The shrinkage coefficients can take values from 0 (no shrinkage) to 1 (complete shrinkage), and the prior on each κ_j has a horseshoe shape. Calculation of the shrinkage coefficients does not affect the statistical results but is helpful for interpretation.

The effective fraction f of nonzero coefficients is

$$f = \frac{1}{J}\sum_{j=1}^{J}(1-\kappa_j)$$

To obtain the prior expectation of f, we compute $E[1-\kappa_j \mid \tau]$ by averaging over the half-Cauchy prior $\lambda_j \sim C^+(0,1)$, which has density $p(\lambda) = \frac{2}{\pi(1+\lambda^2)}$ for $\lambda > 0$. The derivation is exact for the standard horseshoe; for the regularized horseshoe it is a good approximation because the regularization mainly affects the slab (large λ_j) component, while the spike component that determines the degree of shrinkage when τ is small is unchanged.

Setting $a = \tau^2\mathcal{I}_j$:

$$E[1-\kappa_j \mid \tau] = \int_0^\infty \frac{a\lambda^2}{1+a\lambda^2}\cdot\frac{2}{\pi(1+\lambda^2)}\,d\lambda = \frac{2}{\pi}\int_0^\infty \frac{a\lambda^2}{(1+a\lambda^2)(1+\lambda^2)}\,d\lambda$$

We split the integrand:

$$\frac{a\lambda^2}{(1+a\lambda^2)(1+\lambda^2)} = \frac{1}{1+\lambda^2} - \frac{1}{(1+a\lambda^2)(1+\lambda^2)}$$

Partial fractions on the second term (for $a \neq 1$):

$$\frac{1}{(1+a\lambda^2)(1+\lambda^2)} = \frac{1}{a-1}\left[\frac{a}{1+a\lambda^2} - \frac{1}{1+\lambda^2}\right]$$

Integrating over $[0,\infty)$ using $\int_0^\infty \frac{d\lambda}{1+b\lambda^2} = \frac{\pi}{2\sqrt{b}}$:

$$\int_0^\infty \frac{d\lambda}{(1+a\lambda^2)(1+\lambda^2)} = \frac{1}{a-1}\left[\frac{\pi\sqrt{a}}{2} - \frac{\pi}{2}\right] = \frac{\pi}{2}\cdot\frac{\sqrt{a}-1}{a-1} = \frac{\pi}{2(1+\sqrt{a})}$$

where the last step uses $(\sqrt{a}-1)/(a-1) = 1/(1+\sqrt{a})$. Combining:

$$\int_0^\infty \frac{a\lambda^2}{(1+a\lambda^2)(1+\lambda^2)}\,d\lambda = \frac{\pi}{2} - \frac{\pi}{2(1+\sqrt{a})} = \frac{\pi\sqrt{a}}{2(1+\sqrt{a})}$$

Therefore:

$$E[1-\kappa_j \mid \tau] = \frac{2}{\pi}\cdot\frac{\pi\sqrt{a}}{2(1+\sqrt{a})} = \frac{\sqrt{a}}{1+\sqrt{a}} = \frac{\tau\sqrt{\mathcal{I}_j}}{1+\tau\sqrt{\mathcal{I}_j}}$$

The prior expectation of f is then:

$$\mathbb{E}\langle f \mid \tau\rangle = \frac{1}{J}\sum_{j=1}^{J}\frac{\tau\sqrt{\mathcal{I}_j}}{1+\tau\sqrt{\mathcal{I}_j}}$$

The information $\mathcal{I}_j$ about β_j depends on the sample sizes in the studies from which the summary statistics $\hat{\alpha}_j$ and $\hat{\gamma}_j$ were obtained. At $\theta = 0$, $\beta_j = \gamma_j$ and $\mathcal{I}_j$ is asymptotically equivalent to the inverse variance of the maximum likelihood estimate of γ_j, so $\mathcal{I}_j = s_{\gamma(j)}^{-2}$.

We set s_{global} so that the median of the prior on τ equals τ_0, defined as the value satisfying $\mathbb{E}\langle f \mid \tau_0\rangle = f_0$, where f_0 is the prior guess at the fraction of nonzero direct effects.

Assuming equal information $\mathcal{I}_j = \mathcal{I}$ across instruments and solving:

$$f_0 = \frac{\tau_0\sqrt{\mathcal{I}}}{1+\tau_0\sqrt{\mathcal{I}}}$$

Rearranging:

$$f_0\left(1+\tau_0\sqrt{\mathcal{I}}\right) = \tau_0\sqrt{\mathcal{I}} \implies f_0 = (1-f_0)\tau_0\sqrt{\mathcal{I}} \implies \tau_0 = \frac{f_0}{1-f_0}\cdot\frac{1}{\sqrt{\mathcal{I}}}$$

For linear regression with N observations, residual variance σ_y^2, and standardized instruments ($\mathrm{Var}(Z_j) = 1$), the Fisher information is $\mathcal{I} = N/\sigma_y^2$, recovering the formula $\tau_0 = \frac{f_0}{1-f_0}\frac{\sigma_y}{\sqrt{N}}$ of Piironen and Vehtari.

Where the information is not equal across instruments, we can solve numerically for τ_0.

For a general model, Piironen and Vehtari use the term *pseudovariance* for the inverse Fisher information per observation; $1/\sqrt{\mathcal{I}}$ is the corresponding pseudostandard deviation, equal to the standard error of the maximum likelihood estimate of γ_j. With more information (larger $\mathcal{I}$), a smaller τ is required to encode the same prior expectation of f (Piironen and Vehtari 2017).

Extension to summary-level data

Most 2-sample MR studies rely entirely on summary-level data. Even if an individual-level dataset is available, it is computationally faster to use summary statistics from that dataset than to model the individual-level data.

For a Mendelian randomization study of a single exposure with J unlinked genetic instruments, we specify a model with two vector-valued parameters and one scalar parameter:

- $\boldsymbol{\alpha}$ vector of coefficients of effects of the instruments on exposure X,
- $\boldsymbol{\beta}$ vector of coefficients of direct (pleiotropic) effects of the instruments on outcome Y,
- θ causal effect of X on Y.

The crude effect of the jth instrument on the outcome is the sum of the direct effect and the effect through the exposure:

$$\gamma_j = \beta_j + \theta\alpha_j$$

As the instruments are unlinked, we can model the coefficient estimates $\hat{\gamma}_j, \hat{\alpha}_j$ as independent Gaussian variables conditional on the true values:

$$\hat{\alpha}_j \sim \mathcal{N}\left(\alpha_j, s^2_{\alpha(j)}\right)$$

$$\hat{\gamma}_j \sim \mathcal{N}\left(\gamma_j, s^2_{\gamma(j)}\right)$$

With priors on θ and $\boldsymbol{\beta}$, we can specify a full Bayesian probability model, and sample from the joint posterior distribution over the direct effects β_j, the instrument-exposure effects α_j, and the causal effect θ, given the maximum likelihood estimates $\hat{\alpha}_j, \hat{\gamma}_j$ and their standard errors $s_{\gamma_j}, s_{\alpha_j}$.

As in the model for individual-level data, the prior for the direct effects $\beta_1, \ldots, \beta_J$ is

$$\beta_j \sim \mathcal{N}\left(0, \tau^2\tilde{\lambda}_j^2\right), \;\; \tilde{\lambda}_j^2 = \frac{\eta^2\lambda_j^2}{\eta^2 + \tau^2\lambda_j^2}$$

Half-Cauchy priors are specified on the local scale parameters λ_j, and a half-t prior is specified on the global scale parameter τ:

$$\lambda_j \sim C^+\left(0, 1\right), \;\; \tau \sim t^+\left(0, s_{\text{global}}, \nu_{\text{global}}\right)$$

Constructing scalar instruments from multiple SNPs

Standard methods for two-sample Mendelian randomization require as input data an estimate of two scalar quantities for each unlinked instrument: the instrument-exposure coefficient and the instrument-outcome coefficient. We describe below how to combine multiple SNPs from a given region into a single instrument and calculate a scalar coefficient for the effect of this new instrument on the exposure given univariate summary statistics for the association of each SNP with the exposure.

From summary statistics we obtain a table of exposure-associated SNPs with regression coefficients and standard errors, filtered by p-value. After ordering by genomic position, we can group the SNPs into regions such that no gap between adjacent SNPs in the same region exceeds L Mb, and each gap greater than L Mb defines a new region. We use the term "locus" for a region that contains SNPs associated with the exposure at a given p-value threshold. For each locus, we calculate multivariable coefficient estimates $\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}$ by multiplying the vector of univariate coefficient estimates $\hat{\boldsymbol{\alpha}}_{\boldsymbol{u}}$ by the inverse of the correlation matrix $\boldsymbol{\Sigma}_{\boldsymbol{g}}$ between the SNP genotypes.

$$\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}} = \boldsymbol{\Sigma}_{\boldsymbol{g}}^{-\mathbf{1}}\hat{\boldsymbol{\alpha}}_{\boldsymbol{u}}$$

The correlation matrix $\boldsymbol{\Sigma}_{\boldsymbol{g}}$ is obtained from a reference panel such as 1000 Genomes. A shrinkage penalty (equivalent to ridge regression) can be imposed by adding a penalty factor to the diagonal elements of $\boldsymbol{\Sigma}_{\boldsymbol{g}}$. Where the correlation matrix is singular or ill-conditioned, a pseudo-inverse solution is used to calculate the multivariable coefficients. For an individual with genotypes $\boldsymbol{g}$ at the exposure-associated SNPs in a locus, a locus-specific score S predicting the exposure is calculated as $\boldsymbol{g}^\top \cdot \hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}$.

Because the score S is calculated from the genotypes and the genotype-exposure coefficients, we cannot use it as a genotypic instrument for the exposure (the instrument-exposure coefficient is incorporated in the score).

We can however construct a genotypic instrument by factoring the dot product $\boldsymbol{g}^\top \cdot \hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}$ as the product of two scalars (using the definition of the dot product of two Euclidean vectors): the magnitude of the multivariable coefficient vector $\|\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}\|$ and a pseudo-genotype $\|\boldsymbol{g}\|\rho_{\boldsymbol{g},\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}$, where $\rho_{\boldsymbol{g},\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}$ is the correlation between $\boldsymbol{g}$ and $\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}$, geometrically equivalent to the cosine of the angle between these vectors. The pseudo-genotype $\|\boldsymbol{g}\|\rho_{\boldsymbol{g},\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}$ can be thought of as the projection of the genotype vector $\boldsymbol{g}$ on to the exposure effects vector $\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}$, while $\|\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}\|$ is the magnitude of the effect on the exposure. We can then substitute the pseudo-genotype $\|\boldsymbol{g}\|\rho_{\boldsymbol{g},\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}} = S/\|\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}\|$ for the scalar instrument Z and $\|\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}\|$ for the scalar coefficient estimate $\hat{\alpha}$ in the statistical model. The derived coefficient for the estimated effect of the instrument Z on the exposure X is always positive; flipping the coding of the alleles would flip the sign of $\rho_{\boldsymbol{g},\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}$. We can calculate the standard error of $\hat{\alpha}$ as $1/\sqrt{\mathcal{I}_\alpha}$ where $\mathcal{I}_\alpha$ is the Fisher information about the slope of the linear regression of X on Z, given by

$$\mathcal{I}_\alpha = N_\alpha \frac{\mathrm{Var}\,(Z)}{\sigma_X^2 - \hat{\alpha}^2 \mathrm{Var}\,(Z)}$$

where N_α is the sample size of the study from which the estimated univariate coefficients $\hat{\boldsymbol{\alpha}}_{\boldsymbol{u}}$ were obtained, $\mathrm{Var}\,(Z)$ is the variance of the scalar instrument Z estimated in a reference panel, and σ_X^2 is the variance of the exposure X. If an estimate of σ_X^2 is not given with the summary statistics for SNP-exposure associations, it can be calculated from the allele frequency p_j of the jth SNP, the standard error s_{g_j} of the jth univariate coefficient estimate $\hat{\alpha}_{u_j}$, and the sample size N_α as

$$\sigma_X^2 = 2p_j\,(1-p_j)\left(N_\alpha s_{g_j}^2 + \hat{\alpha}_{u_j}^2\right)$$

This procedure ensures that all exposure-associated SNPs can be used in constructing the genotypic instruments, and that these instruments are unlinked so that their pleiotropic effects can be modelled as independent.

Estimation of instrument-outcome coefficients from summary level data

Where an individual-level dataset for genotype-outcome associations is available, the instrument-outcome coefficients for scalar instruments constructed from multiple SNPs can be estimated by fitting regression models for the effect of each instrument on the outcome. Where only summary-level data are available for the genotype-outcome associations, it is possible to obtain an estimate $\hat{\gamma}$ of the coefficient for the effect of the scalar instrument Z on the outcome and its standard error, using individual-level genotype data from a reference panel of individual with similar ancestry to that in the genotype-outcome dataset, such as 1000 Genomes. Although we have not used this method, we give a brief derivation for completeness. First, the vector $\hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}$ of estimated multivariable coefficients is calculated by premultiplying the estimates $\hat{\boldsymbol{\gamma}}_{\boldsymbol{u}}$ of the univariate coefficients by $\boldsymbol{\Sigma}_{\boldsymbol{G}}^{-\boldsymbol{1}}$. The coefficient $\hat{\gamma}$ for the scalar instrument Z is then estimated by minimizing the sum of the squared differences between the predictor calculated from Z and the linear predictor calculated from the vector of genotypes $\boldsymbol{G}$, where the sum is taken over the genotypes of all n individuals in the reference panel.

$$\sum_{i=1}^{n} \left(\gamma Z_i - \boldsymbol{G}_i \cdot \hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}\right)^2$$

Equating to zero the derivative of this expression with respect to γ, substituting $\|\boldsymbol{G}_{\boldsymbol{i}}\|\rho_{G,\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}$ for Z_i and factoring the dot product $\boldsymbol{G}_i \cdot \hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}$ as $\|\boldsymbol{G}_i\|\rho_{\boldsymbol{G}_i,\hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}}\|\hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}\|$ we obtain

$$\hat{\gamma} = \|\hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}\| \frac{\sum_{i=1}^{n} \rho_{\boldsymbol{G}_i,\hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}} \|\boldsymbol{G}_i\|^2 \rho_{G_i,\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}}{\sum_{i=1}^{n} \|\boldsymbol{G}_{\boldsymbol{i}}\|^2 \rho_{G_i,\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}^2}$$

The ratio in this expression can be recognized as a weighted average of the ratio $\rho_{G_i,\hat{\boldsymbol{\gamma}}_{\boldsymbol{m}}}/\rho_{G_i,\hat{\boldsymbol{\alpha}}_{\boldsymbol{m}}}$, with weights Z_i^2. The standard error of $\hat{\gamma}$ is $1/\sqrt{\mathcal{I}_\gamma}$ where $\mathcal{I}_\gamma$ is the Fisher information on the slope of the regression of Y on Z, given (for a logistic regression model) by

$$\mathcal{I}_\gamma = N_\gamma \mathrm{Var}\left(Z\right) p\left(1-p\right)$$

where N_γ and p are respectively the total sample size and the proportion of cases in the dataset from which the coefficient estimates $\hat{\boldsymbol{\gamma}}_{\boldsymbol{u}}$ were obtained.

References

Burgess, Stephen, Christopher N. Foley, Elias Allara, James R. Staley, and Joanna M. M. Howson. 2020. "A Robust and Efficient Method for Mendelian Randomization with Hundreds of Genetic Variants." *Nat Commun* 11 (1): 376. https://doi.org/10.1038/s41467-019-14156-4.

Carvalho, Carlos M., Nicholas G. Polson, and James G. Scott. 2009. "Handling Sparsity via the Horseshoe." In *Proc. Twelth Int. Conf. Artif. Intell. Stat.*, 73–80. PMLR.

Piironen, Juho, and Aki Vehtari. 2017. "Sparsity Information and Regularization in the Horseshoe and Other Shrinkage Priors." *Electron. J. Stat.* 11: 5018–51.

Xue, Haoran, Xiaotong Shen, and Wei Pan. 2021. "Constrained Maximum Likelihood-Based Mendelian Randomization Robust to Both Correlated and Uncorrelated Pleiotropic Effects." *Am J Hum Genet* 108 (7): 1251–69. https://doi.org/10.1016/j.ajhg.2021.05.014.